# Digital Signal Processing

## Spectral Estimation Techniques II

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# Periodogram

MS spectrum $S_{ms}[k]$ unit is Watts

$$\hat{S}_{ms}[k] = \left| \frac{FFT\{x_N[n]\}}{N} \right|^2$$

We want the density in watts per hertz

Frequency bin width: $\Delta f = \frac{f_s}{\text{nperseg}}$

$$\sum \hat{S}_{ms}[k] = \sum \hat{S}[k] \cdot \Delta f$$

W

W/hz

hz

$$\hat{S}(f) \approx \hat{S}[k] = \frac{\hat{S}_{ms}[k]}{\Delta f} = \frac{N}{F_s} \hat{S}_{ms}[k]$$

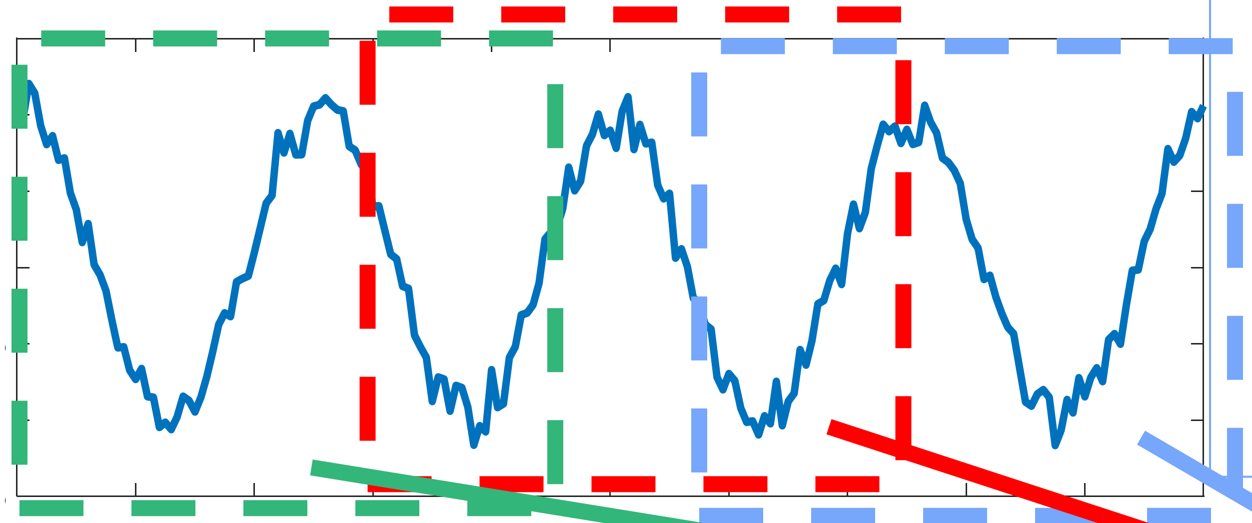
Periodogram definition

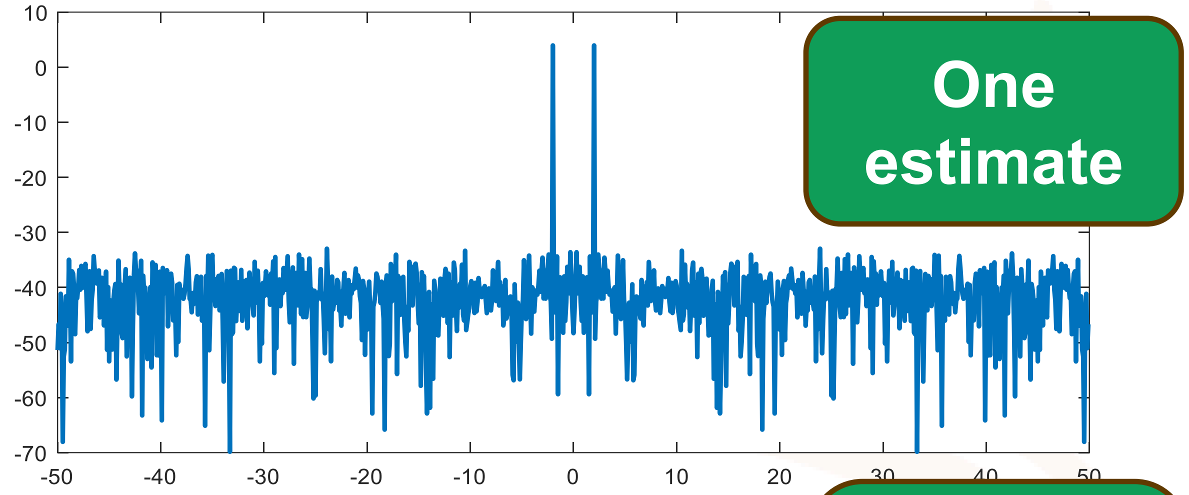
$$\hat{S}(f) \approx \hat{S}[k] = \frac{1}{N\,BW} |FFT\{x_N[n]\}|^2$$

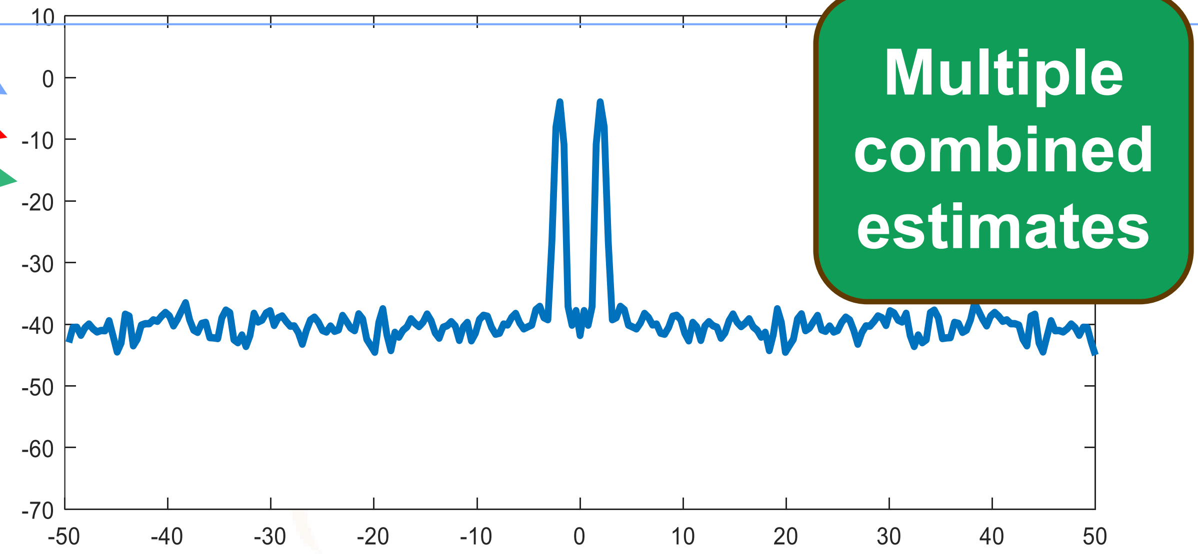
Since the periodogram does not learn parameters, it is a non-parametric estimator

# Welch: averages of periodograms

**One periodogram is one estimate/measurement of $S(f)$ of a random process**

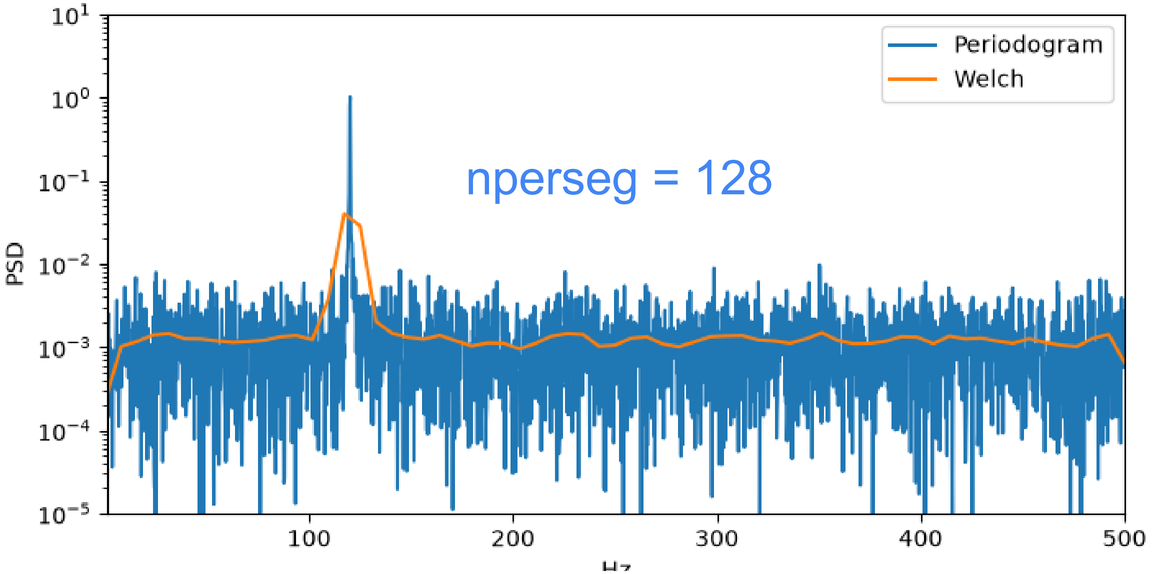


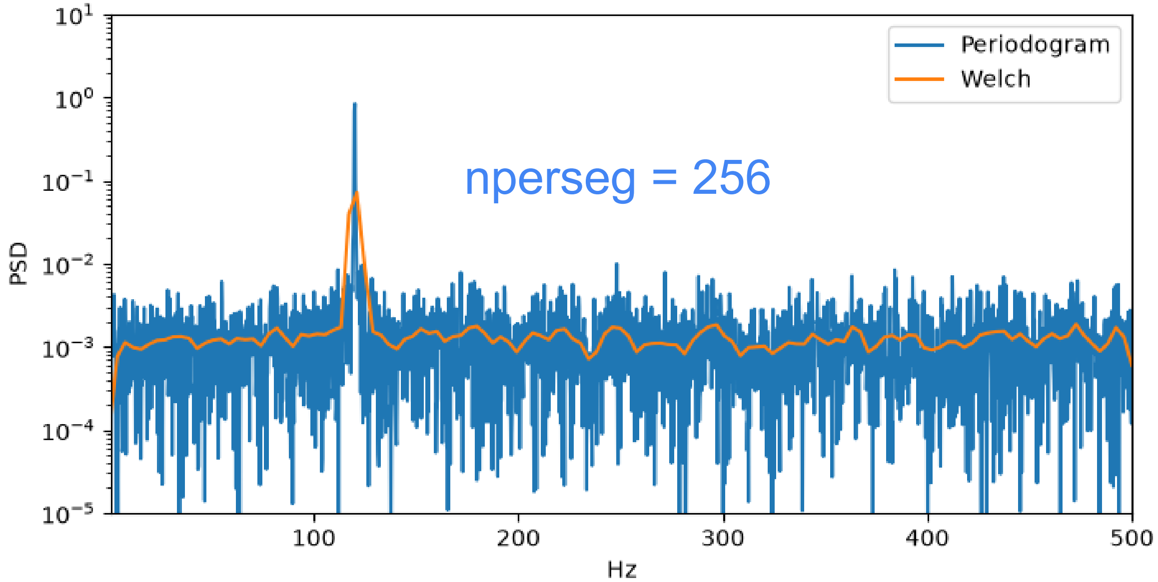


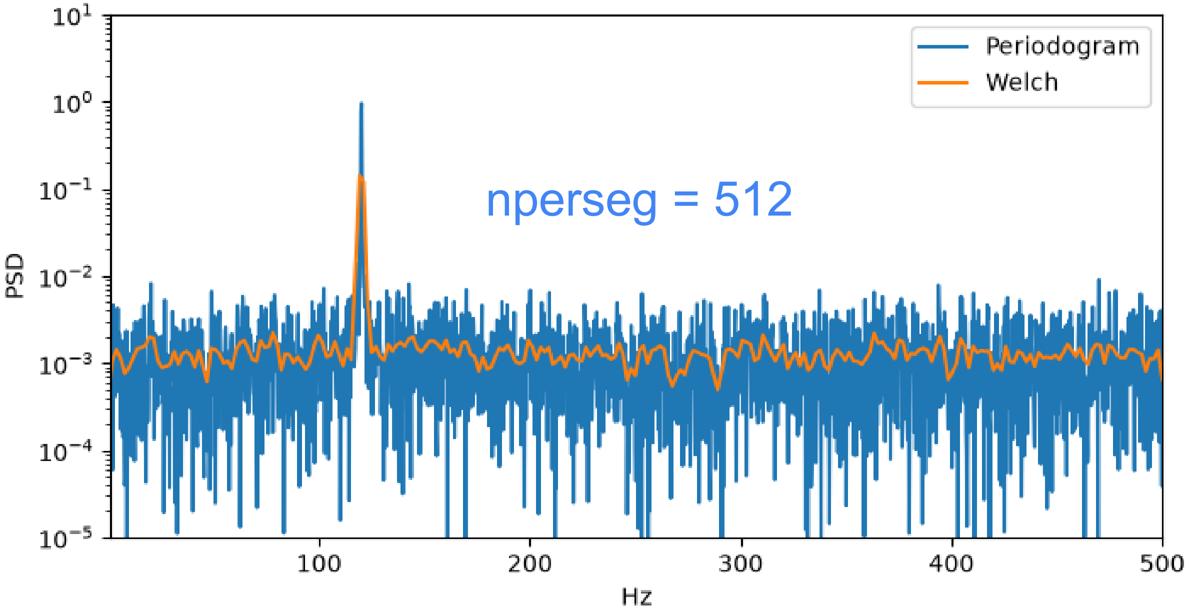



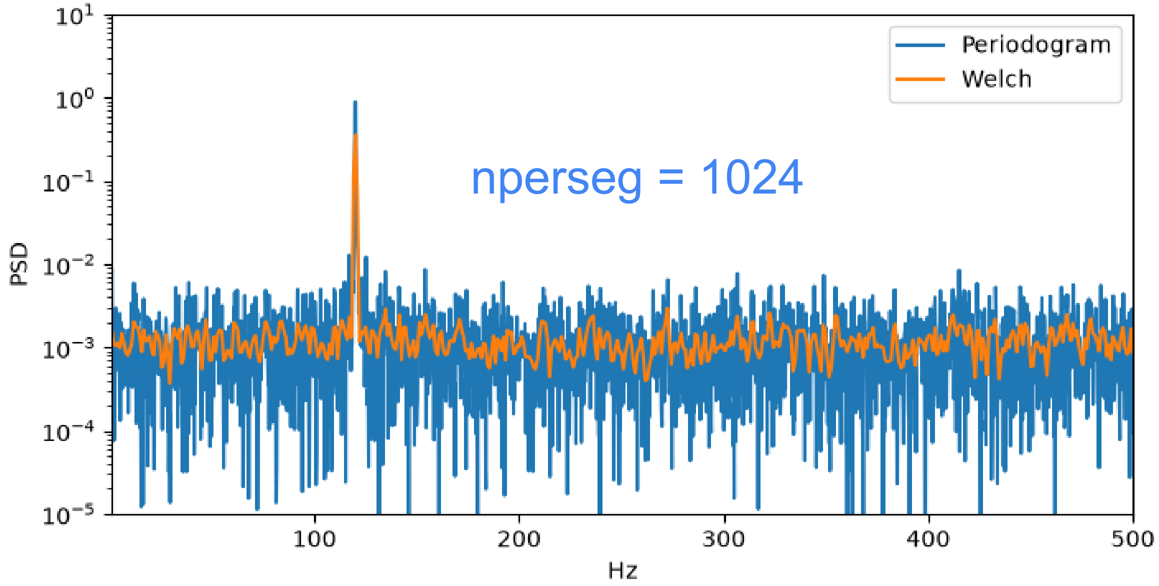

**Welch's improves variance by averaging the power over shorter windows, sacrificing frequency resolution**

Frequency bin width: $\Delta f = \dfrac{f_s}{\text{nperseg}}$

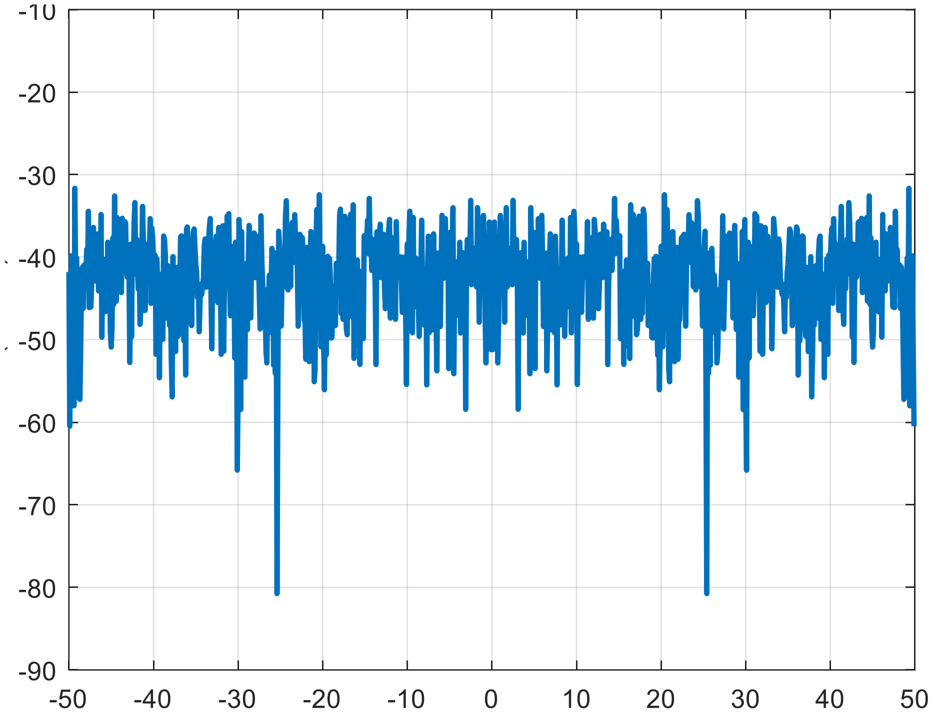


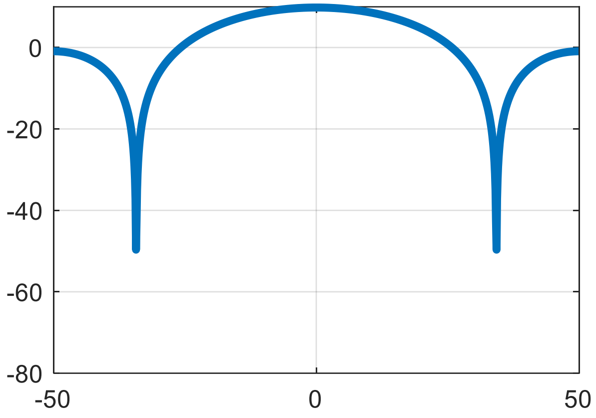



# Filtering white noise shapes the noise spectrum like the filter

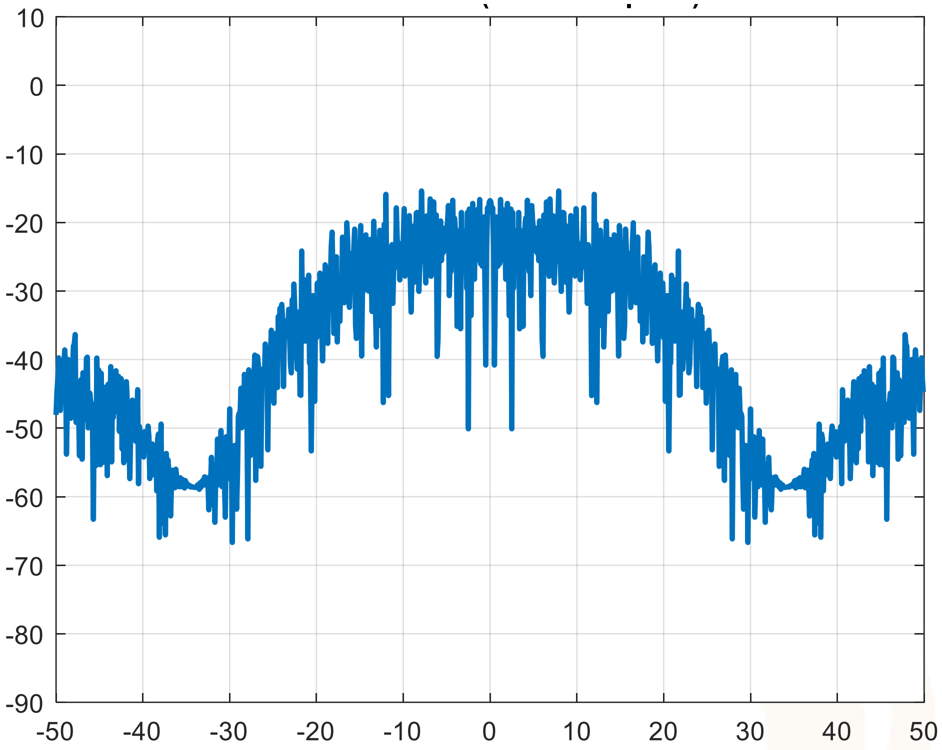



$\hat{S}_x(f)$

**White noise**




$|H(f)|^2$




$\hat{S}_y(f) = |H(f)|^2 \hat{S}_x(f)$

**Colored noise**

# Autoregressive (AR) Modeling

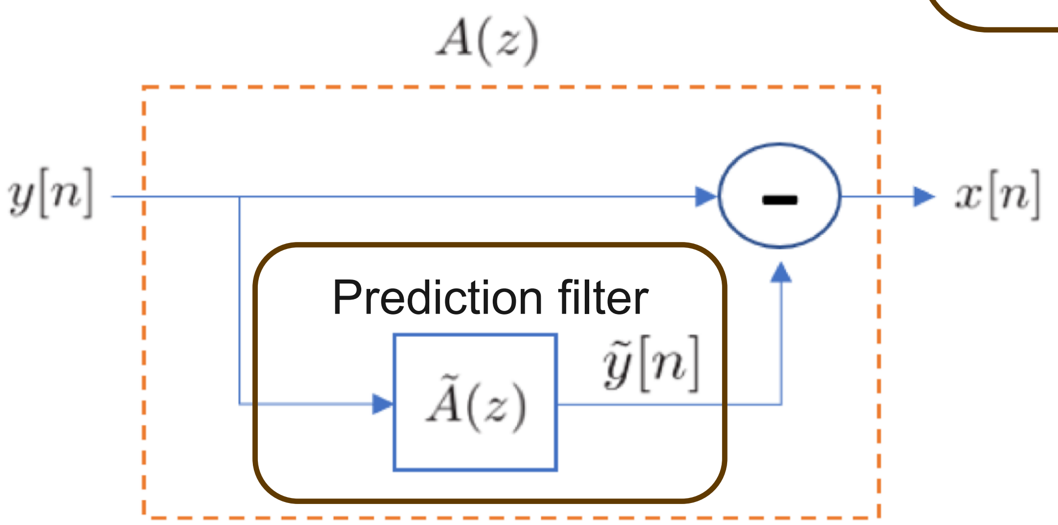
**White Noise** $x[n]$ → $H(z)$ → **Colored noise** $y[n]$ → $A(z)$ → **White Noise** $\tilde{x}[n]$

**Model assumption:** $H(z)$ **is all-poles**

**Collected signal**

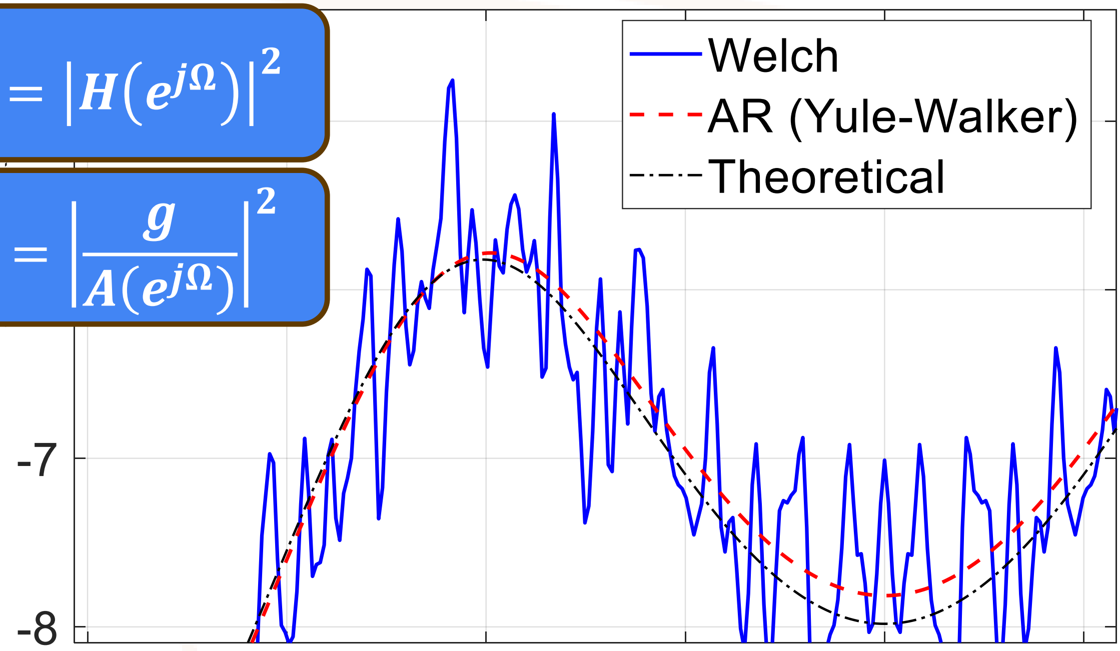
**Find coefficients of FIR filter with $a_0 = 1$ that minimizes output energy**

**Optimum $A(z)$ (Yule-Walker) yields white noise with power $g^2$**



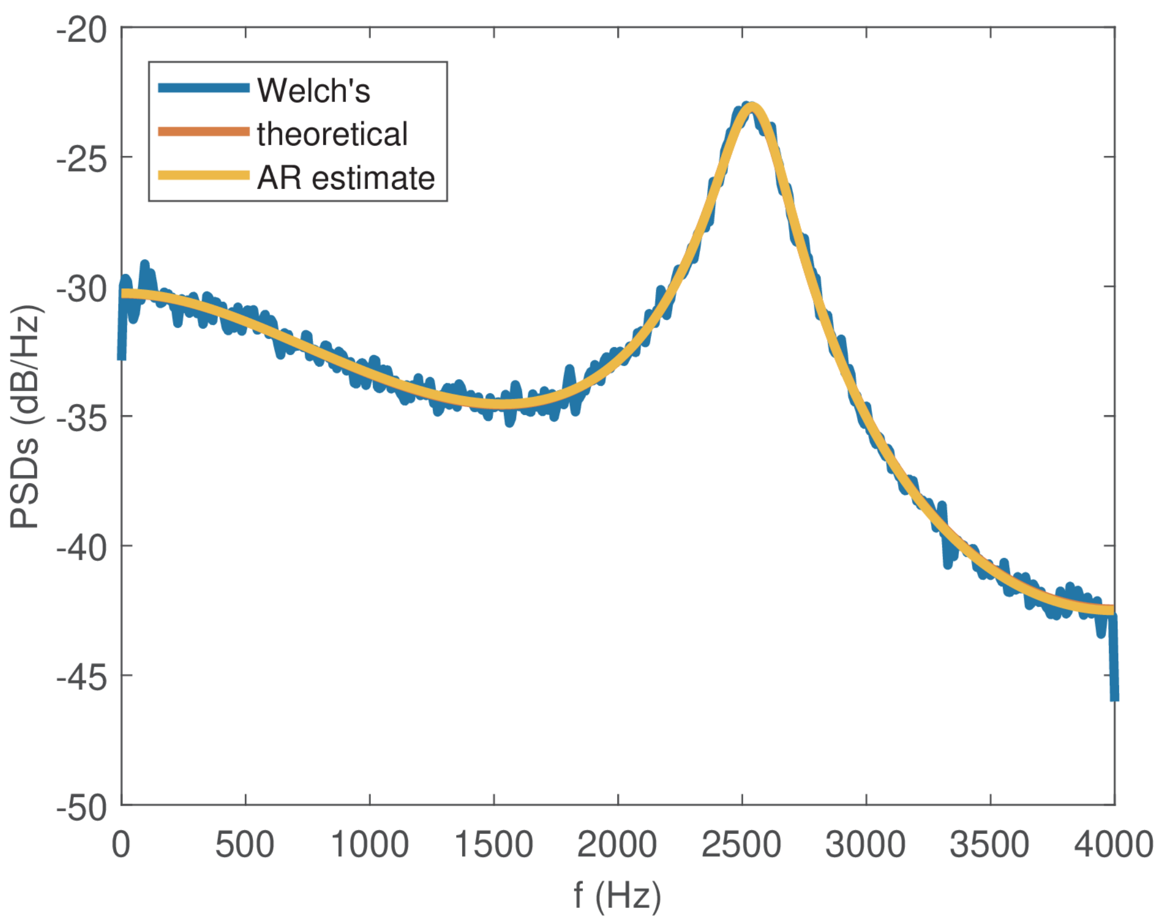

$$S(e^{j\Omega}) = \left|H(e^{j\Omega})\right|^2$$

$$\hat{S}(e^{j\Omega}) = \left|\frac{g}{A(e^{j\Omega})}\right|^2$$



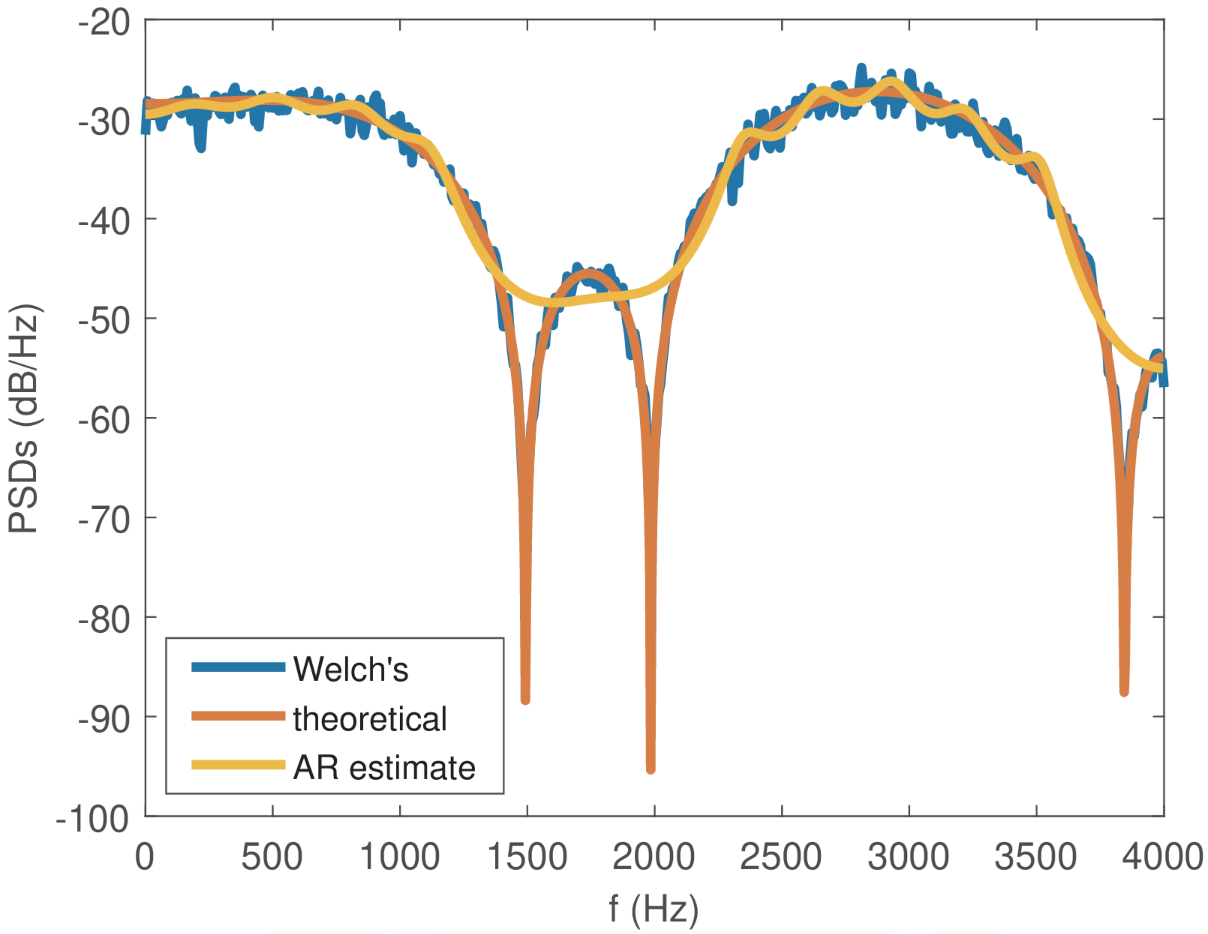

**AR model is widely used in speech coding to mimic voice spectrum**

## Satisfying the assumptions of the AR model




**Far fewer coefficients compared to Welch's method**

## Not satisfying the assumptions of the AR model (system has zeros)

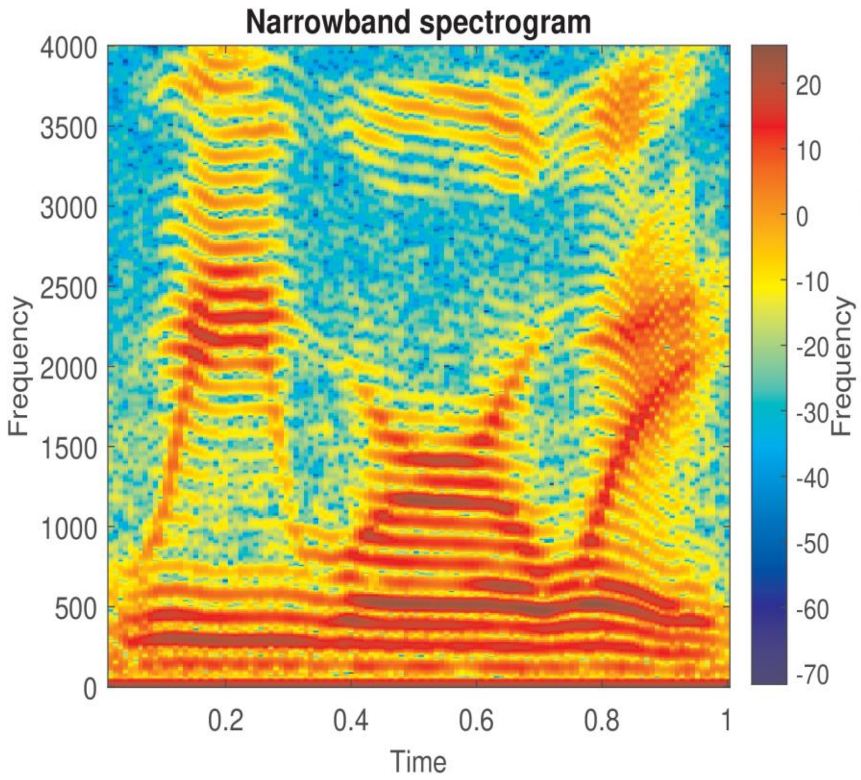



**Poles cannot create zeros, so AR model performs poorly near notches**

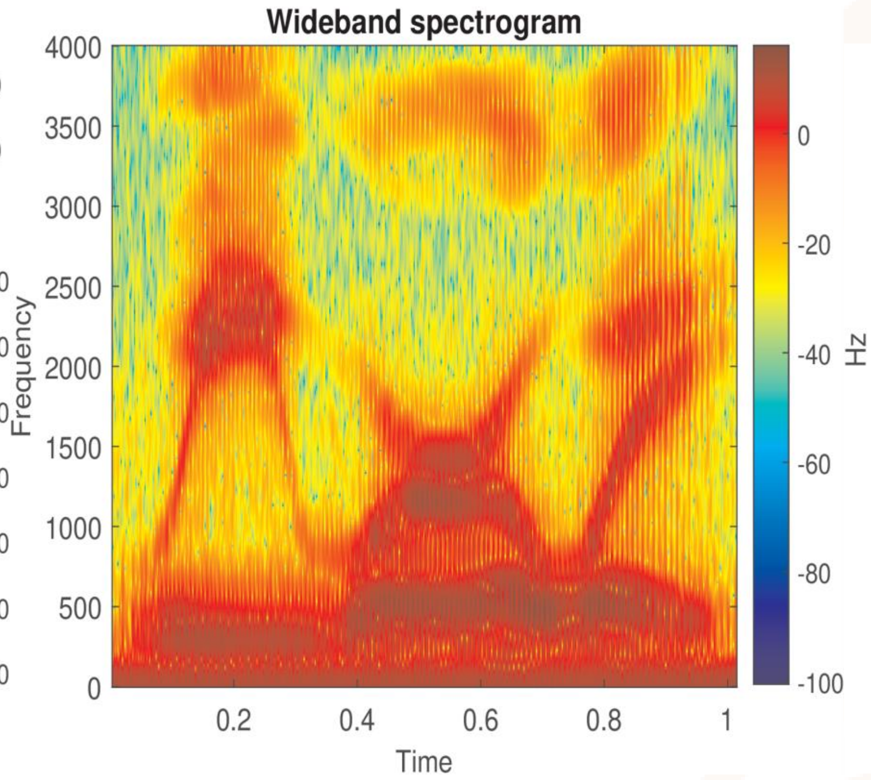
# Spectrogram

**Nonstationary processes have time-varying spectrum**

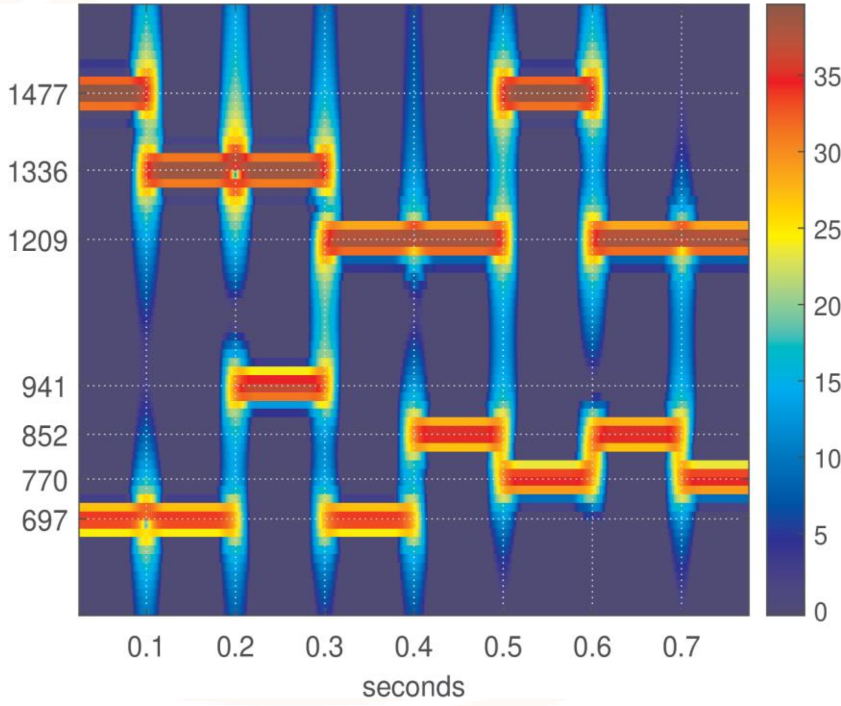
**Spectrograms take multiple windows but do not combine them like Welch's method**




**Speech**




**Speech**




**Piano**

# Thank you!

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